

Qualification Pack



Pipeline Fitter

Electives: Carry out Industrial Pipe Fitting/ Carry out Municipal Pipe Fitting

QP Code: PSC/Q0107

Version: 2.0

NSQF Level: 3.5

Water Management and Plumbing Skill Council || Unit No-606, DLF Prime Tower
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PSC/Q0107: Pipeline Fitter

Brief Job Description

A Pipeline Fitter specializes in the installation, repair, and maintenance of pipelines used for various purposes, such as transporting liquids and gases. The individual carries out pipe fitting using the appropriate tools and equipment as per the applicable blueprints, drawings and specifications.

Personal Attributes

The individual should be able to work independently and be comfortable in performing laborious work. The person should be able to work as per the applicable requirements and in coordination with others while maintaining a positive attitude. The individual should have appropriate verbal and written communication skills.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSC/N0161: Install pipelines for internal and external distribution systems](#)
2. [PSC/N0162: Carry out Repair and Maintenance of different types of Pipelines](#)
3. [PSC/N0136: Apply Health and Safety Practices at the Workplace](#)
4. [PSC/N0137: Work Effectively with Others](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Electives (mandatory to select at least one):

Elective 1: Carry out Industrial Pipe Fitting

This unit deals in detail with laying and installing internal and basic external pipes for both, internal and external distribution systems and their accessories.

1. [PSC/N0185: Carry out Industrial Pipe Fitting](#)

Elective 2: Carry out Municipal Pipe Fitting

This unit deals in detail with laying and installing internal and basic external pipes for both, internal and external distribution systems and their accessories.

1. [PSC/N0186: Carry out Municipal Pipe Fitting](#)

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Qualification Pack (QP) Parameters

Sector	Plumbing
Sub-Sector	Industrial / Non-Industrial Plumbing
Occupation	Plumbing Systems Installation and Maintenance
Country	India
NSQF Level	3.5
Credits	15
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ 7126.0201, NCO-2015/ 7126.0301
Minimum Educational Qualification & Experience	11th grade pass OR 8th grade pass with 1.5 years of experience relevant experience in plumbing operations OR Previous relevant Qualification of NSQF Level 2.5 with 3 Years of experience relevant experience in plumbing operations OR Previous relevant Qualification of NSQF Level (NSQF Level 3) with 1.5 years of experience relevant experience in plumbing operations
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	27/08/2027
NSQF Approval Date	27/08/2024
Version	2.0
Reference code on NQR	QG-3.5-PL-02941-2024-V1-WMPS
NQR Version	2.0

Remarks:



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NIL

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PSC/N0161: Install pipelines for internal and external distribution systems

Description

This unit deals in detail with laying and installing internal and basic external pipes for both, internal and external distribution systems and their accessories.

Scope

The scope covers the following :

- Prepare for installation
- Install the pipes
- Carry out post-installation activities

Elements and Performance Criteria

Prepare for pipeline installation

To be competent, the user/individual on the job must be able to:

- PC1.** study blueprints, drawings and specifications to determine the type of pipes and tools to be used
- PC2.** conduct a route survey
- PC3.** determine the fitting requirements to install the pipes
- PC4.** create detailed sketches for pipe and equipment fabrication and installation, as required
- PC5.** . plan the sequence of tasks
- PC6.** prepare the area for installation
- PC7.** transport and stack the pipes, fittings, and other materials at the site
- PC8.** measure, cut, thread, groove, bend, and assemble metal, plastic and fibreglass pipes, valves and fittings using couplings, clamps, screws, bolts, etc.
- PC9.** carry out cutting, caulking and threading appropriately for effective joining of pipes
- PC10.** establish the sequence of internal and external pipe installations
- PC11.** plan installation in or around obstructions such as electrical wiring, pavements, etc.
- PC12.** locate and mark the position of pipe installations, connections, passage holes, and fixtures in structures, using measuring instruments such as rulers and levels
- PC13.** . prepare trenches and ditches for housing the pipelines
- PC14.** cut openings in structures to accommodate pipes and pipe fittings, using hand and power tools (if required)
- PC15.** measure, cut, thread, and bend pipe to the required angle, using hand and power tools or equipment such as cutting, threading and bending equipment

Install the pipeline

To be competent, the user/individual on the job must be able to:

- PC16.** install horizontal and vertical supports to hold pipes in place
- PC17.** string pipes along the trench

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- PC18.** bend pipes in the field to suit the topography of the site
- PC19.** apply coating and wrapping to steel pipes
- PC20.** join pipe sections and related equipment and secure them in position
- PC21.** fix the valves
- PC22.** adhere to external levels and standard benchmarks concerning sewerage and drainage
- PC23.** achieve quality finishing in the repair of piping systems and fittings
- PC24.** ensure timely completion of work and minimal wastage of material and damage to other systems

Carry out post-installation activities

To be competent, the user/individual on the job must be able to:

- PC25.** connect piping network with water supply and sewerage lines
- PC26.** examine the pipe for any welding flaws or leakage at the joints
- PC27.** appropriate testing equipment to piping systems for leaks
- PC28.** cover the trenches by soil and restoration of the land to its original appearance
- PC29.** clear the work area and dispose of waste

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of pipelines classified on the basis of manufacturing material, transported substance, and function of the pipes
- KU2.** complete knowledge of establishing levels and slopes of drainage and sewerage piping
- KU3.** basic plumbing terminology
- KU4.** pipeline construction phases
- KU5.** route survey purpose and process
- KU6.** ditching or trenching process
- KU7.** distinguishing between levels and distances of water and sewer lines
- KU8.** service line networks such as water and sewer mains and its connection norms
- KU9.** techniques related to cutting, bending and joining of pipes and fittings
- KU10.** various types of defects such as leakages, improper alignment, etc.
- KU11.** considerations while transporting and stacking the pipes, fittings, and other materials
- KU12.** purpose and process for stringing the pipes along the trench
- KU13.** tools and techniques for bending pipes in the field to suit topography of the site
- KU14.** tools and techniques for joining pipes before or after lowering into the trench
- KU15.** coating and wrapping of pipes, the need, types, tools and materials
- KU16.** valve fixing process
- KU17.** horizontal and vertical pipe supporting systems
- KU18.** pipeline examination process for welding flaws or leakage at the joints
- KU19.** purpose and process of covering trenches by soil and restoration of the land to its original appearance

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- KU20.** units and measurements
- KU21.** safety precautions
- KU22.** response to emergencies e.g., water supply failures, fire and system failures
- KU23.** use of measuring instruments e.g., steel scale, steel tape, micrometer, Vernier, etc.
- KU24.** compilation of pressure test results in prescribed format

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare checklists, reports and fill out forms in local language or Hindi/English
- GS2.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS3.** read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or Hindi
- GS4.** state information, doubts and concerns about work related matters in local language or Hindi/English
- GS5.** plan one's daily tasks to achieve maximum productivity
- GS6.** establish priorities and deadlines in consultation with others and record them
- GS7.** be punctual and work as per agreed priorities
- GS8.** manage distractions and maintain workplace discipline
- GS9.** listen to customer's concerns and doubts carefully and address them
- GS10.** be courteous
- GS11.** identify ways to increase productivity and reduce errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for pipeline installation</i>	17	23	-	10
PC1. study blueprints, drawings and specifications to determine the type of pipes and tools to be used	-	-	-	-
PC2. conduct a route survey	-	-	-	-
PC3. determine the fitting requirements to install the pipes	-	-	-	-
PC4. create detailed sketches for pipe and equipment fabrication and installation, as required	-	-	-	-
PC5. . plan the sequence of tasks	-	-	-	-
PC6. prepare the area for installation	-	-	-	-
PC7. transport and stack the pipes, fittings, and other materials at the site	-	-	-	-
PC8. measure, cut, thread, groove, bend, and assemble metal, plastic and fibreglass pipes, valves and fittings using couplings, clamps, screws, bolts, etc.	-	-	-	-
PC9. carry out cutting, caulking and threading appropriately for effective joining of pipes	-	-	-	-
PC10. establish the sequence of internal and external pipe installations	-	-	-	-
PC11. plan installation in or around obstructions such as electrical wiring, pavements, etc.	-	-	-	-
PC12. locate and mark the position of pipe installations, connections, passage holes, and fixtures in structures, using measuring instruments such as rulers and levels	-	-	-	-
PC13. . prepare trenches and ditches for housing the pipelines	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. cut openings in structures to accommodate pipes and pipe fittings, using hand and power tools (if required)	-	-	-	-
PC15. measure, cut, thread, and bend pipe to the required angle, using hand and power tools or equipment such as cutting, threading and bending equipment	-	-	-	-
<i>Install the pipeline</i>	9	19	-	7
PC16. install horizontal and vertical supports to hold pipes in place	-	-	-	-
PC17. string pipes along the trench	-	-	-	-
PC18. bend pipes in the field to suit the topography of the site	-	-	-	-
PC19. apply coating and wrapping to steel pipes	-	-	-	-
PC20. join pipe sections and related equipment and secure them in position	-	-	-	-
PC21. fix the valves	-	-	-	-
PC22. adhere to external levels and standard benchmarks concerning sewerage and drainage	-	-	-	-
PC23. achieve quality finishing in the repair of piping systems and fittings	-	-	-	-
PC24. ensure timely completion of work and minimal wastage of material and damage to other systems	-	-	-	-
<i>Carry out post-installation activities</i>	4	8	-	3
PC25. connect piping network with water supply and sewerage lines	-	-	-	-
PC26. examine the pipe for any welding flaws or leakage at the joints	-	-	-	-
PC27. appropriate testing equipment to piping systems for leaks	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. cover the trenches by soil and restoration of the land to its original appearance	-	-	-	-
PC29. clear the work area and dispose of waste	-	-	-	-
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0161
NOS Name	Install pipelines for internal and external distribution systems
Sector	Plumbing
Sub-Sector	Industrial / Non-Industrial Plumbing
Occupation	Plumbing Systems Installation and Maintenance
NSQF Level	3.5
Credits	6
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

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PSC/N0162: Carry out Repair and Maintenance of different types of Pipelines

Description

This unit is about the activities that involve the repair and maintenance of various types of pipelines.

Scope

The scope covers the following :

- Prepare for pipeline repair work
- Repair the pipeline
- Carry out post-repair work
- Carry out maintenance of pipeline

Elements and Performance Criteria

Prepare for pipeline repair work

To be competent, the user/individual on the job must be able to:

- PC1.** Interpret the pipeline layout drawing and identify the pipeline job requirements
- PC2.** identify the specifications of the pipeline from the layout drawings
- PC3.** identify the problem/fault in the system and its cause
- PC4.** diagnose the reason for failure
- PC5.** establish the sequence of repair/maintenance
- PC6.** assemble fittings, fixtures and tools required for the task

Repair the pipeline

To be competent, the user/individual on the job must be able to:

- PC7.** isolate the repair area
- PC8.** locate and mark the position of the component to be replaced/repared, using measuring instruments such as rulers and levels
- PC9.** cut openings in structures/trenches to remove defected pipes and pipe fittings, using hand and power tools
- PC10.** measure, cut, thread, and bend pipes to required angle, using hand and power tools or equipment such as cutting, threading and bending equipment
- PC11.** remove and replace worn pipe assemblies, components and fittings, and carry out general maintenance work
- PC12.** re-assemble the pipeline with fittings post correcting the defect
- PC13.** repair internal piping network connections with external service mains such as water and sewerage lines

Carry out post-repair work

To be competent, the user/individual on the job must be able to:

- PC14.** test the pipe joints and fixtures for proper functioning
- PC15.** use the appropriate testing equipment to check systems for leaks

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PC16. handover to authorities detailed documentation of faults, their causes, and the repair work carried out

PC17. clear the work area and waste disposal

Carry out maintenance of pipeline

To be competent, the user/individual on the job must be able to:

PC18. dismantle the pipeline circuit, as needed

PC19. fasten mechanical components/ subassemblies together using pipe, pipe fittings using hand/ power tools

PC20. set and adjust the pipeline / fittings using a pipe spanner and hand tools

PC21. ensure the layout of pipelines as per specified circuit drawings

PC22. check proper clamping to ensure the rigidity

PC23. perform pressure test of pipeline to ensure it is leak-proof before handover

PC24. record the test results in the prescribed format of the organization

PC25. supervise tapping of pipelines by private plumbers for domestic, commercial/industrial connections

PC26. respond to emergencies concerning pipelines as per standard operating procedures

PC27. ensure all activities are complete according to the maintenance checklist

PC28. document maintenance carried out, deviation from the checklist and observations about the given job

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. complete knowledge of establishing levels and slopes of drainage and sewerage piping

KU2. basic plumbing terminology

KU3. pipeline lay out / circuit drawings

KU4. distinguish between levels and distances of water and sewer lines

KU5. service line networks such as water and sewer mains and its connection norms

KU6. various types of pipelines and their specifications

KU7. techniques related to cutting, bending and joining of fittings and fixtures

KU8. various types of defects such as leakages, improper alignment, etc.

KU9. tools and tackles with specified fittings to be used for the job

KU10. common problems and faults that occur in pipelines

KU11. activities regarding repair of various pipeline faults

KU12. implications of not adhering to sequence of activities and operations

KU13. safety precautions to be taken while working on a particular gas/steam/high pressure lines etc.

KU14. assembling techniques such as aligning, bending, fixing, mechanical jointing,

KU15. threaded jointing, sealing etc.

KU16. steps required to assemble/ dismantle pipeline circuit as per pipeline lay out

KU17. checks that need to be made to ensure that the pipeline circuit is safe and ready

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- KU18.** measurements that are performed pertaining to plumbing systems installation
- KU19.** relevant hand and power tools such as wrenches, plier, screwdriver, power drill, pipe cutter, crimping tool, pipe bender, threading tool, hacksaw, metal file, etc.
- KU20.** general response procedures to emergencies such as water supply failures, fire and system failures
- KU21.** purpose and process of tapping of pipeline by private plumbers for domestic, commercial/industrial connections
- KU22.** compilation of pressure test results in prescribed format
- KU23.** units of measurement
- KU24.** material disposal procedure, importance of appropriate disposal of material

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare checklists, reports and fill out forms in local language or Hindi/English
- GS2.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS3.** read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or Hindi
- GS4.** state information, doubts and concerns about work related matters in local language or Hindi/English
- GS5.** plan one's daily tasks to achieve maximum productivity
- GS6.** establish priorities and deadlines in consultation with others and record them
- GS7.** be punctual and work as per agreed priorities
- GS8.** manage distractions and maintain workplace discipline
- GS9.** listen to customer's concerns and doubts carefully and address them
- GS10.** be courteous
- GS11.** identify ways to increase productivity and reduce errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for pipeline repair work</i>	7	9	-	6
PC1. Interpret the pipeline layout drawing and identify the pipeline job requirements	-	-	-	-
PC2. identify the specifications of the pipeline from the layout drawings	-	-	-	-
PC3. identify the problem/fault in the system and its cause	-	-	-	-
PC4. diagnose the reason for failure	-	-	-	-
PC5. establish the sequence of repair/maintenance	-	-	-	-
PC6. assemble fittings, fixtures and tools required for the task	-	-	-	-
<i>Repair the pipeline</i>	12	20	-	8
PC7. isolate the repair area	-	-	-	-
PC8. locate and mark the position of the component to be replaced/repared, using measuring instruments such as rulers and levels	-	-	-	-
PC9. cut openings in structures/trenches to remove defected pipes and pipe fittings, using hand and power tools	-	-	-	-
PC10. measure, cut, thread, and bend pipes to required angle, using hand and power tools or equipment such as cutting, threading and bending equipment	-	-	-	-
PC11. remove and replace worn pipe assemblies, components and fittings, and carry out general maintenance work	-	-	-	-
PC12. re-assemble the pipeline with fittings post correcting the defect	-	-	-	-
PC13. repair internal piping network connections with external service mains such as water and sewerage lines	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out post-repair work</i>	4	10	-	3
PC14. test the pipe joints and fixtures for proper functioning	-	-	-	-
PC15. use the appropriate testing equipment to check systems for leaks	-	-	-	-
PC16. handover to authorities detailed documentation of faults, their causes, and the repair work carried out	-	-	-	-
PC17. clear the work area and waste disposal	-	-	-	-
<i>Carry out maintenance of pipeline</i>	7	11	-	3
PC18. dismantle the pipeline circuit, as needed	-	-	-	-
PC19. fasten mechanical components/ subassemblies together using pipe, pipe fittings using hand/ power tools	-	-	-	-
PC20. set and adjust the pipeline / fittings using a pipe spanner and hand tools	-	-	-	-
PC21. ensure the layout of pipelines as per specified circuit drawings	-	-	-	-
PC22. check proper clamping to ensure the rigidity	-	-	-	-
PC23. perform pressure test of pipeline to ensure it is leak-proof before handover	-	-	-	-
PC24. record the test results in the prescribed format of the organization	-	-	-	-
PC25. supervise tapping of pipelines by private plumbers for domestic, commercial/industrial connections	-	-	-	-
PC26. respond to emergencies concerning pipelines as per standard operating procedures	-	-	-	-
PC27. ensure all activities are complete according to the maintenance checklist	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. document maintenance carried out, deviation from the checklist and observations about the given job	-	-	-	-
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0162
NOS Name	Carry out Repair and Maintenance of different types of Pipelines
Sector	Plumbing
Sub-Sector	Industrial / Non-Industrial Plumbing
Occupation	Plumbing Systems Installation and Maintenance
NSQF Level	3.5
Credits	5
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

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PSC/N0136: Apply Health and Safety Practices at the Workplace

Description

This unit is about practices to be executed at the workplace which is crucial for creating a safe and secure working environment that promotes the well-being of employees and minimizes the risk of accidents, injuries, and occupational illnesses. Here's a description outlining the key aspects of implementing health and safety practices at the workplace.

Scope

The scope covers the following :

- Follow safety measures to avoid accidents
- Daily inspection of tools and equipment
- Deal with emergencies

Elements and Performance Criteria

Follow safety measures to avoid accidents

To be competent, the user/individual on the job must be able to:

- PC1.** identify risks and hazards at the workplace
- PC2.** wear personal protective equipment (PPE) as per the type of plumbing work
- PC3.** place protective barricades and signages around the pits and trenches
- PC4.** isolate the plumbing fittings and fixtures from electrical wiring to avoid accidents
- PC5.** adhere to organisational procedures for reporting hazards and incidents to relevant authorities
- PC6.** establish ventilation before entering underground work areas
- PC7.** work safely in and around trenches, elevated places and confined areas
- PC8.** ensure tools and hazardous materials are not left unattended
- PC9.** ensure good housekeeping in order to prevent hazards e.g. fire
- PC10.** dispose waste materials and used PPE according to regulations and codes of practice

Follow hygiene and sanitation practices

To be competent, the user/individual on the job must be able to:

- PC11.** follow recommended personal hygiene and sanitation practices, for example, washing/sanitizing hands, covering face with a bent elbow while coughing/sneezing etc.
- PC12.** clean and disinfect work area, materials/supplies, equipment etc. before and after use.
- PC13.** report hygiene and sanitation issues to appropriate authority

Use tools, equipment and materials safely

To be competent, the user/individual on the job must be able to:

- PC14.** check that the tools, equipment and materials are in good condition and as per industry standards before use
- PC15.** use power tools and machinery that are grounded
- PC16.** replace or repair split or loose tools before use

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PC17. store and transport various plumbing materials safely

Deal with emergencies

To be competent, the user/individual on the job must be able to:

PC18. follow workplace emergency and evacuation procedures

PC19. use a fire extinguisher correctly

PC20. use safe methods to free a person from electrocution

PC21. administer appropriate first aid (such as CPR etc.) to victims in case of cuts, bleeding, burns, choking, electric shock, poisoning etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. possible causes of risk, hazard or accident in the workplace

KU2. organisational procedures for upkeep of tools and plumbing materials, health and safety

KU3. location of all the general health and safety equipment in the workplace

KU4. meaning of hazards and risks

KU5. hazardous environment encountered during work such as underground areas, elevated areas, areas with water and electricity supply, presence of biological waste, under construction sites etc.

KU6. work practices and precautions to control and prevent risks, hazards and accidents

KU7. importance of each personal protective equipment used such as eye protection mask, hard hats, gloves, apron, rubber boots etc.

KU8. tools and plumbing equipment as per latest industry standards

KU9. preventative and remedial actions to be taken in case of exposure to toxic materials

KU10. specific safety and health related problems faced in domestic, commercial and institutional setups

KU11. various causes of fire and precautionary activities to prevent the fire accident

KU12. techniques of using the different fire extinguishers

KU13. rescue techniques applied during a fire hazard

KU14. various types of safety signs and meaning

KU15. appropriate basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, minor burns, poisoning, eye injuries etc.

KU16. potential injuries and ill health associated with incorrect handling of tools and equipment

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. write an accident/incident report in local language or English

GS2. read and comprehend labels, charts, signages, manuals, plumbing symbols etc.

GS3. question coworkers appropriately in order to clarify instructions and other issues



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- GS4.** plan and organize the work schedule, work area, tools, equipment and materials for improved productivity
- GS5.** determine key considerations and priorities when faced with problems
- GS6.** seek official and authorised sources of help and guidance to resolve problems that cannot be solved at one's level of authority

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow safety measures to avoid accidents</i>	12	32	-	5
PC1. identify risks and hazards at the workplace	1	2	-	0.5
PC2. wear personal protective equipment (PPE) as per the type of plumbing work	1	2	-	0.5
PC3. place protective barricades and signages around the pits and trenches	1	4	-	0.5
PC4. isolate the plumbing fittings and fixtures from electrical wiring to avoid accidents	1	4	-	0.5
PC5. adhere to organisational procedures for reporting hazards and incidents to relevant authorities	1	4	-	0.5
PC6. establish ventilation before entering underground work areas	1	4	-	0.5
PC7. work safely in and around trenches, elevated places and confined areas	1	3	-	0.5
PC8. ensure tools and hazardous materials are not left unattended	1	3	-	0.5
PC9. ensure good housekeeping in order to prevent hazards e.g. fire	2	3	-	0.5
PC10. dispose waste materials and used PPE according to regulations and codes of practice	2	3	-	0.5
<i>Follow hygiene and sanitation practices</i>	3	8	-	2
PC11. follow recommended personal hygiene and sanitation practices, for example, washing/sanitizing hands, covering face with a bent elbow while coughing/sneezing etc.	1	3	-	1
PC12. clean and disinfect work area, materials/supplies, equipment etc. before and after use.	1	3	-	0.5
PC13. report hygiene and sanitation issues to appropriate authority	1	2	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Use tools, equipment and materials safely</i>	7	8	-	1
PC14. check that the tools, equipment and materials are in good condition and as per industry standards before use	2	2	-	0.25
PC15. use power tools and machinery that are grounded	2	2	-	0.25
PC16. replace or repair split or loose tools before use	1	2	-	0.25
PC17. store and transport various plumbing materials safely	2	2	-	0.25
<i>Deal with emergencies</i>	8	12	-	2
PC18. follow workplace emergency and evacuation procedures	2	3	-	0.5
PC19. use a fire extinguisher correctly	2	3	-	0.5
PC20. use safe methods to free a person from electrocution	2	3	-	0.5
PC21. administer appropriate first aid (such as CPR etc.) to victims in case of cuts, bleeding, burns, choking, electric shock, poisoning etc.	2	3	-	0.5
NOS Total	30	60	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0136
NOS Name	Apply Health and Safety Practices at the Workplace
Sector	Plumbing
Sub-Sector	Industrial / Non-Industrial Plumbing, Water Supply & Water Treatment & Quality Control, Sewerage & Sewage Treatment, Drainage, Water Harvesting & Ground Recharging, Firefighting & Safety Systems, Gas & Piping (Industrial / Medical), HVAC & Steam, Manufacturing of Plumbing / Firefighting Products, Micro-Hydel Power
Occupation	Plumbing Systems Installation and Maintenance, Designing, Sales, Material Handling, Plumbing Servicing, Plumbing Plant Operations
NSQF Level	4
Credits	0.5
Version	2.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQF Clearance Date	18/02/2025

Qualification Pack

PSC/N0137: Work Effectively with Others

Description

This unit covers basic etiquette and competencies required in behavior and interactions with others at the workplace.

Scope

The scope covers the following :

- Communicate effectively
- Work in a team effectively
- Respect diversity

Elements and Performance Criteria

Communicate effectively

To be competent, the user/individual on the job must be able to:

- PC1.** obtain complete information and instructions
- PC2.** seek clarifications from appropriate source when required
- PC3.** provide information accurately and clearly
- PC4.** use inclusive language (verbal, non-verbal and written) that is gender, disability and culturally sensitive

Work in a team effectively

To be competent, the user/individual on the job must be able to:

- PC5.** prioritize tasks as required
- PC6.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC7.** escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict
- PC8.** act upon constructive feedback from others

Respect diversity

To be competent, the user/individual on the job must be able to:

- PC9.** transact with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture
- PC10.** recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace
- PC11.** report incidents of harassment and discrimination to appropriate authority

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to employment, behavior, harassment, discrimination and performance conditions

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- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** escalation matrix and procedures for reporting work and employment related issues
- KU4.** types of harassment and discrimination based on gender, disability, caste, religion and culture and how to recognize it.
- KU5.** importance of effective communication and the impact of poor communication on the employee, the employer and the customer
- KU6.** importance of teamwork in organizational and individual success
- KU7.** various components of effective communication such as tone and pitch
- KU8.** importance of ethics and discipline for professional success
- KU9.** how to express and address grievances appropriately and effectively
- KU10.** importance and ways of managing interpersonal conflict effectively
- KU11.** different types of disabilities and the challenges faced by persons with disability (PwD)
- KU12.** laws, acts and provisions defined for PwD by the statutory bodies
- KU13.** government and private schemes and benefits available for PwD
- KU14.** Importance of gender sensitivity and equality.
- KU15.** gender, disability and cultural biases, stereotypes and impact on others
- KU16.** gender and its concepts such as gender roles, gender spectrum, gender as an identity
- KU17.** legislations, grievance redressal mechanisms, and penalties against harassment in the workplace

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write clear and legible notes, keep records, prepare to-do lists and take down instructions
- GS2.** write basic numbers, quantities and basic work-related terminology for operational requirements in the local language and English
- GS3.** read basic terminologies to accurately interpret labels, supervisor's instructions in the local language and English
- GS4.** read and interpret accurate information from work-related documents and various relevant work instructions and records in local
- GS5.** interact with the concerned personnel appropriately (correct protocol and manner of speaking etc.)
- GS6.** display active listening skills while interacting with co-workers and others in the workplace
- GS7.** deliver consistent and reliable service to internal and external customers
- GS8.** work with co-workers and supervisor to resolve any issues that threaten work quality as per the planned schedule

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate effectively</i>	11	26	-	2
PC1. obtain complete information and instructions	3	8	-	-
PC2. seek clarifications from appropriate source when required	1	5	-	-
PC3. provide information accurately and clearly	3	7	-	1
PC4. use inclusive language (verbal, non-verbal and written) that is gender, disability and culturally sensitive	4	6	-	1
<i>Work in a team effectively</i>	11	25	-	3
PC5. prioritize tasks as required	3	8	-	1
PC6. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	3	8	-	1
PC7. escalate grievances and problems to appropriate authority as per procedure to resolve them and avoid conflict	3	4	-	1
PC8. act upon constructive feedback from others	2	5	-	-
<i>Respect diversity</i>	8	12	-	2
PC9. transact with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture	3	5	-	1
PC10. recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace	3	5	-	1
PC11. report incidents of harassment and discrimination to appropriate authority	2	2	-	-
NOS Total	30	63	-	7

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0137
NOS Name	Work Effectively with Others
Sector	Plumbing
Sub-Sector	Industrial / Non-Industrial Plumbing, Water Supply & Water Treatment & Quality Control, Sewerage & Sewage Treatment, Drainage, Water Harvesting & Ground Recharging, Firefighting & Safety Systems, Gas & Piping (Industrial / Medical), HVAC & Steam, Manufacturing of Plumbing / Firefighting Products, Micro-Hydel Power
Occupation	Plumbing Systems Installation and Maintenance, Designing, Sales, Material Handling, Plumbing Servicing, Plumbing Plant Operations
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQF Clearance Date	27/08/2024

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Qualification Pack

PSC/N0185: Carry out Industrial Pipe Fitting

Description

This unit deals in detail with laying and installing internal and basic external pipes for both, internal and external distribution systems and their accessories.

Scope

The scope covers the following :

- Prepare for industrial pipe fitting activities
- Install the industrial pipe fittings
- Inspect, test, and commission the piping system

Elements and Performance Criteria

Prepare for industrial pipe fitting activities

To be competent, the user/individual on the job must be able to:

- PC1.** evaluate the project requirements, including the type of fluid, pressure, temperature, and environmental conditions
- PC2.** study the basic drawings and schematics of the piping system, to determine the pipe routes, materials, and dimensions
- PC3.** conduct a thorough inspection of the site to identify potential obstacles and ensure the environment is suitable for installation
- PC4.** use appropriate materials based on the project requirements, such as steel, copper, PVC, etc.
- PC5.** measure and cut pipes to the required lengths using tools like pipe cutters, saws, or torches.
- PC6.** assemble sections of the piping system in a controlled environment before transporting them to the installation site.

Install the industrial pipe fittings

To be competent, the user/individual on the job must be able to:

- PC7.** position the pipes correctly according to the schematics, ensuring proper alignment to prevent leaks and ensure smooth fluid flow
- PC8.** join pipes using appropriate joining methods, such as welding, threading, and flanging, depending on the type of pipe and the specific application.
- PC9.** install supports and anchors to secure the pipes and prevent movement due to pressure or thermal expansion

Inspect, test and commission the piping system

To be competent, the user/individual on the job must be able to:

- PC10.** check the piping system visually for any obvious defects, misalignments, or improper joints.
- PC11.** perform pressure tests to ensure the system can handle the designated pressure without leaking.
- PC12.** carry out Non-Destructive Testing (NDT) following appropriate methods, such as Xrays and ultrasonic testing to inspect welds and joints for internal defects
- PC13.** carry out appropriate repair and maintenance of the piping system based on test results.

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- PC14.** clean the piping system to remove debris, contaminants, and residues from the installation process.
- PC15.** conduct final checks to ensure all components are installed correctly and functioning as intended.
- PC16.** test the system under normal operating conditions to verify performance and ensure there are no leaks or other issues.
- PC17.** maintain the appropriate records concerning installation, inspections, and tests in compliance with industry standards and regulations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the industrial pipe fitting requirements
- KU2.** different types of pipes, e.g., steel, copper, Polyvinyl Chloride (PVC) and High-density Polyethylene (HDPE), and their applications
- KU3.** the properties of different material properties, such as strength, corrosion resistance, thermal expansion, and chemical compatibility
- KU4.** how to read and interpret blueprints, schematics, and isometric drawings to understand the layout and specifications of the piping system
- KU5.** the industry-standard symbols and notations used in piping diagrams
- KU6.** how to take accurate measurements and calculate lengths, angles, and dimensions
- KU7.** how to perform basic calculations for pressure, flow rates, and pipe sizing
- KU8.** various welding techniques, e.g., arc welding, Metal Inert Gas (MIG) welding and Tungsten Inert Gas (TIG) welding
- KU9.** the threading and flanging methods for joining pipes
- KU10.** the proper use of different types of fittings and couplings
- KU11.** the use of relevant hand and power tools, e.g. pipe wrenches, cutters, grinders, saws, etc.
- KU12.** the use of measuring instruments, such as calipers, micrometers, and levels
- KU13.** the use of appropriate protective equipment to protect against hazards in pipe fitting activities
- KU14.** the principles of fluid flow, including laminar and turbulent flow, pressure drops, and flow rates
- KU15.** how to design piping systems to ensure efficient and reliable fluid transport
- KU16.** the visual inspection methods and non-destructive testing (NDT) techniques, such as X-rays, ultrasonic testing, and magnetic particle testing
- KU17.** how to conduct hydrostatic and pneumatic pressure tests to verify the integrity of the piping system
- KU18.** the process of diagnosing and troubleshooting issues in piping systems
- KU19.** various repair methods, including patching, pipe replacement, and joint repair
- KU20.** the relevant industry codes and standards
- KU21.** the relevant documentation requirements

Generic Skills (GS)



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User/individual on the job needs to know how to:

- GS1.** identify solutions to work-related issues
- GS2.** maintain work-related notes and records
- GS3.** communicate clearly and politely
- GS4.** perform basic calculations
- GS5.** read the relevant guides and literature to get the latest information about the field of work
- GS6.** identify solutions to work-related issues
- GS7.** plan and priorities tasks to ensure timely completion
- GS8.** make quick decisions in case of an emergency/ accident

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for industrial pipe fitting activities</i>	14	20	-	9
PC1. evaluate the project requirements, including the type of fluid, pressure, temperature, and environmental conditions	-	-	-	-
PC2. study the basic drawings and schematics of the piping system, to determine the pipe routes, materials, and dimensions	-	-	-	-
PC3. conduct a thorough inspection of the site to identify potential obstacles and ensure the environment is suitable for installation	-	-	-	-
PC4. use appropriate materials based on the project requirements, such as steel, copper, PVC, etc.	-	-	-	-
PC5. measure and cut pipes to the required lengths using tools like pipe cutters, saws, or torches.	-	-	-	-
PC6. assemble sections of the piping system in a controlled environment before transporting them to the installation site.	-	-	-	-
<i>Install the industrial pipe fittings</i>	9	15	-	5
PC7. position the pipes correctly according to the schematics, ensuring proper alignment to prevent leaks and ensure smooth fluid flow	-	-	-	-
PC8. join pipes using appropriate joining methods, such as welding, threading, and flanging, depending on the type of pipe and the specific application.	-	-	-	-
PC9. install supports and anchors to secure the pipes and prevent movement due to pressure or thermal expansion	-	-	-	-
<i>Inspect, test and commission the piping system</i>	7	15	-	6
PC10. check the piping system visually for any obvious defects, misalignments, or improper joints.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. perform pressure tests to ensure the system can handle the designated pressure without leaking.	-	-	-	-
PC12. carry out Non-Destructive Testing (NDT) following appropriate methods, such as Xrays and ultrasonic testing to inspect welds and joints for internal defects	-	-	-	-
PC13. carry out appropriate repair and maintenance of the piping system based on test results.	-	-	-	-
PC14. clean the piping system to remove debris, contaminants, and residues from the installation process.	-	-	-	-
PC15. conduct final checks to ensure all components are installed correctly and functioning as intended.	-	-	-	-
PC16. test the system under normal operating conditions to verify performance and ensure there are no leaks or other issues.	-	-	-	-
PC17. maintain the appropriate records concerning installation, inspections, and tests in compliance with industry standards and regulations	-	-	-	-
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0185
NOS Name	Carry out Industrial Pipe Fitting
Sector	Plumbing
Sub-Sector	
Occupation	Plumbing Systems Installation and Maintenance
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

PSC/N0186: Carry out Municipal Pipe Fitting

Description

This unit deals in detail with laying and installing internal and basic external pipes for both, internal and external distribution systems and their accessories.

Scope

The scope covers the following :

- Prepare for municipal pipe fitting activities
- Install the municipal pipe fittings
- Inspect and test the municipal piping system

Elements and Performance Criteria

Prepare for municipal pipe fitting activities

To be competent, the user/individual on the job must be able to:

- PC1.** determine the project requirements, including the type of fluid (e.g. water and sewage), pressure, and volume
- PC2.** conduct a survey and create maps of the target area for the installation of pipes
- PC3.** ensure the site environment is suitable for pipe fitting
- PC4.** study the detailed plans and schematics for the piping system, specifying pipe routes, materials, and dimensions
- PC5.** check for necessary permits and approvals for pipe fitting from local authorities and regulatory bodies
- PC6.** identify potential obstacles at the installation site and take appropriate preventive measures
- PC7.** check for the availability of the necessary materials, such as pipes, fittings, and valves, pipe fitting operations
- PC8.** coordinate the excavation of trenches along the planned pipe routes, ensuring they are deep and wide enough to accommodate the pipes and any necessary bedding materials
- PC9.** backfill the trenches with soil, compacting it in layers to ensure stability

Install the municipal pipe fittings

To be competent, the user/individual on the job must be able to:

- PC10.** lay pipes in the excavated trenches according to the design plans, ensuring proper alignment and slope to facilitate fluid flow
- PC11.** join pipes using appropriate methods, such as welding, threading, flanging, or using mechanical couplings and fittings
- PC12.** ensure all joints are secure and leak-proof
- PC13.** place appropriate bedding material, such as sand and gravel around the pipes to provide support and prevent damage

Inspect and test the municipal piping system

To be competent, the user/individual on the job must be able to:

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- PC14.** conduct a visual inspection of the entire system to check for any obvious defects or misalignments
- PC15.** perform hydrostatic testing or pneumatic testing to ensure the system can handle the designated pressure without leaking
- PC16.** identify potential leaks in the system using the relevant techniques such as dye testing or electronic leak detection
- PC17.** implement appropriate measures to plug the identified leakages
- PC18.** flush the piping system to remove any debris, contaminants, or residues from the installation process
- PC19.** disinfect water supply systems to ensure they are free from harmful bacteria and pathogens
- PC20.** conduct final checks under normal operating conditions to ensure all components are installed correctly and functioning as intended without leakages or other issues
- PC21.** schedule regular inspections to identify potential issues early and ensure the system continues to operate efficiently
- PC22.** implement appropriate plans for responding to emergencies, such as pipe bursts or leaks, to minimize disruptions to services
- PC23.** maintain appropriate records concerning pipe installation, inspections, and tests

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the industrial pipe fitting requirements
- KU2.** different types of pipes, e.g., steel, copper, Polyvinyl Chloride (PVC) and High-density Polyethylene (HDPE), and their applications
- KU3.** the properties of different material properties, such as strength, corrosion resistance, thermal expansion, and chemical compatibility
- KU4.** how to read and interpret blueprints, schematics, and isometric drawings to understand the layout and specifications of the piping system
- KU5.** the industry-standard symbols and notations used in piping diagrams
- KU6.** how to take accurate measurements and calculate lengths, angles, and dimensions
- KU7.** to perform basic calculations for pressure, flow rates, and pipe sizing
- KU8.** various welding techniques, e.g., arc welding, Metal Inert Gas (MIG) welding and Tungsten Inert Gas (TIG) welding
- KU9.** the threading and flanging methods for joining pipes
- KU10.** the proper use of different types of fittings and couplings
- KU11.** the use of relevant hand and power tools, e.g. pipe wrenches, cutters, grinders, saws, etc.
- KU12.** the use of measuring instruments, such as calipers, micrometers, and levels
- KU13.** the use of appropriate protective equipment to protect against hazards in pipe fitting activities
- KU14.** the principles of fluid flow, including laminar and turbulent flow, pressure drops, and flow rates
- KU15.** how to design piping systems to ensure efficient and reliable fluid transport

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- KU16.** the visual inspection methods and non-destructive testing (NDT) techniques, such as X-rays, ultrasonic testing, and magnetic particle testing
- KU17.** how to conduct hydrostatic and pneumatic pressure tests to verify the integrity of the piping system
- KU18.** the process of diagnosing and troubleshooting issues in piping systems
- KU19.** various repair methods, including patching, pipe replacement, and joint repair
- KU20.** the relevant industry codes and standards
- KU21.** the relevant documentation requirements

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** read the relevant guides and literature to get the latest information about the field of work
- GS3.** communicate clearly and politely
- GS4.** perform basic calculations
- GS5.** listen attentively to understand the instructions being given
- GS6.** identify solutions to work-related issues
- GS7.** plan and prioritise tasks to ensure timely completion
- GS8.** make quick decisions in case of an emergency/ accident

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for municipal pipe fitting activities</i>	12	20	-	9
PC1. determine the project requirements, including the type of fluid (e.g. water and sewage), pressure, and volume	-	-	-	-
PC2. conduct a survey and create maps of the target area for the installation of pipes	-	-	-	-
PC3. ensure the site environment is suitable for pipe fitting	-	-	-	-
PC4. study the detailed plans and schematics for the piping system, specifying pipe routes, materials, and dimensions	-	-	-	-
PC5. check for necessary permits and approvals for pipe fitting from local authorities and regulatory bodies	-	-	-	-
PC6. identify potential obstacles at the installation site and take appropriate preventive measures	-	-	-	-
PC7. check for the availability of the necessary materials, such as pipes, fittings, and valves, pipe fitting operations	-	-	-	-
PC8. coordinate the excavation of trenches along the planned pipe routes, ensuring they are deep and wide enough to accommodate the pipes and any necessary bedding materials	-	-	-	-
PC9. backfill the trenches with soil, compacting it in layers to ensure stability	-	-	-	-
<i>Install the municipal pipe fittings</i>	9	15	-	6
PC10. lay pipes in the excavated trenches according to the design plans, ensuring proper alignment and slope to facilitate fluid flow	-	-	-	-
PC11. join pipes using appropriate methods, such as welding, threading, flanging, or using mechanical couplings and fittings	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. ensure all joints are secure and leak-proof	-	-	-	-
PC13. place appropriate bedding material, such as sand and gravel around the pipes to provide support and prevent damage	-	-	-	-
<i>Inspect and test the municipal piping system</i>	9	15	-	5
PC14. conduct a visual inspection of the entire system to check for any obvious defects or misalignments	-	-	-	-
PC15. perform hydrostatic testing or pneumatic testing to ensure the system can handle the designated pressure without leaking	-	-	-	-
PC16. identify potential leaks in the system using the relevant techniques such as dye testing or electronic leak detection	-	-	-	-
PC17. implement appropriate measures to plug the identified leakages	-	-	-	-
PC18. flush the piping system to remove any debris, contaminants, or residues from the installation process	-	-	-	-
PC19. disinfect water supply systems to ensure they are free from harmful bacteria and pathogens	-	-	-	-
PC20. conduct final checks under normal operating conditions to ensure all components are installed correctly and functioning as intended without leakages or other issues	-	-	-	-
PC21. schedule regular inspections to identify potential issues early and ensure the system continues to operate efficiently	-	-	-	-
PC22. implement appropriate plans for responding to emergencies, such as pipe bursts or leaks, to minimize disruptions to services	-	-	-	-
PC23. maintain appropriate records concerning pipe installation, inspections, and tests	-	-	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0186
NOS Name	Carry out Municipal Pipe Fitting
Sector	Plumbing
Sub-Sector	
Occupation	Plumbing Systems Installation and Maintenance
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSC/N0161.Install pipelines for internal and external distribution systems	30	50	0	20	100	20
PSC/N0162.Carry out Repair and Maintenance of different types of Pipelines	30	50	0	20	100	20
PSC/N0136.Apply Health and Safety Practices at the Workplace	30	60	-	10	100	15
PSC/N0137.Work Effectively with Others	30	63	-	7	100	15
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	140	253	-	57	450	80

Elective: 1 Carry out Industrial Pipe Fitting

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSC/N0185.Carry out Industrial Pipe Fitting	30	50	0	20	100	20
Total	30	50	-	20	100	20

Elective: 2 Carry out Municipal Pipe Fitting

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSC/N0186.Carry out Municipal Pipe Fitting	30	50	0	20	100	20
Total	30	50	-	20	100	20

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NCVET	National Council for Vocational Education and Training
WMPSC	Water Management and Plumbing Skill Council
QP	Qualification Pack
MC	Model Curriculum
NSQF	National Skills Qualification Framework
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards
NCO	National Classification of Occupations
ES	Employability Skills
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
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